Khayra Sarra Zerguerras

Final-Year AI & Data Science Engineering Student

Research Interests: AI Safety, LLM Security, AI Agent Security, Trustworthy AI

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Seeking a PFE / Master's 2 Thesis Research Opportunity – January 2027

Education

École Supérieure en Informatique (ESI-SBA) *2022–2027*
Engineering Degree in AI & Data Science – Expected 2027
Final year of engineering cycle

Baccalaureate in Mathematics *2022*
Graduated with honors – **17.36/20**

Technical Skills

AI & ML	Machine Learning, Deep Learning, NLP, Generative AI, LLMs, Diffusion Models, Conditional GANs, MLOps	**Programming**	Python, SQL, Java, JavaScript, TypeScript, Dart
Data Engineering	PySpark, Apache Spark, Hadoop, Apache Airflow, ETL, Data Lakes, Stream Processing, GraphFrames	**Data & Analytics**	PostgreSQL, MySQL, MongoDB, Data Warehousing, SQL OLAP, PCA, Power BI
Development	Streamlit, React, Next.js, Flutter, Django, Node.js, Express.js, Spring Boot, HTML5, CSS3	**Cloud & Tools**	AWS (EC2, S3), Docker, Git, Postman, Figma

Selected Projects

Indirect Prompt Injection & Tool-Use Guardrail Sandbox *2026–Present*
- Building a controlled research sandbox to evaluate indirect prompt injection attacks against tool-using LLM agents and compare baseline behavior with defensive guardrails.
- Implementing constrained tool access and security policies for untrusted documents, web content, and simulated side effects; designing tests to evaluate defense effectiveness.

Malicious URL Detector *2026*
- Built and compared machine learning classifiers for detecting malicious URLs associated with phishing, malware, spam, and defacement.

Sign Language Production – Text-to-Video *2026*
- Developed a text-to-sign-language generation pipeline using 3D skeletal poses and diffusion models.
- Designed transformer-based text conditioning with cross-attention and contrastive alignment, without an intermediate gloss representation.

Smart Notes Generator – Audio-to-Impact *2026*
- Developed an AI application that transforms audio into structured notes and concise summaries using NLP and language models.

Automatic Image Colorization *2026*
- Implemented a Conditional GAN for automatic grayscale image colorization.

AI-Powered Job Skills Gap Analyzer *2025*
- Developed a Streamlit ML application to analyze gaps between user skills and job requirements.

Market Basket Analysis & Product Recommendation *2026*
- Built a product recommendation system using association rule mining and an MLOps workflow for reproducible development and deployment.

Experience & Activities

Remote Research Collaboration with Dr. Yuxia Wang (INSAIT) *2026–Present*
Research Contribution – Multilingual Harmful Humor Detection and Explanation
- Contributing to an ongoing research project through data annotation for multilingual harmful humor detection and explanation.

CNR (Caisse Nationale des Retraites) *Summer 2025*
Intern – Software Development Department
- Analyzed internal and external applications and documented recommendations for improving internal software and data systems.

Alphabit Club – Finance Department *2024–Present*
- Managed event budgets and prepared financial reports.

Ingenuims Club – Marketing / Visual Creation Department *2022–2024*
- Created visual content and photography for student events.

Certifications

Oracle Certified Generative AI Professional | **AWS Concepts** | **Cloud Computing**

Languages

Arabic (Native) | French (B2) | English (C2 – EF SET, 79/100)